

ALFIYA ANSARI

+4915217428155

 alfiya-ansari175

 alfiya-ansari-175

 Koblenz, Germany

 alfiyaansari175@gmail.com

 alfiya-ansari-175.github.io



SUMMARY

Data Science graduate student with the hands-on experience in data analysis, machine learning, and statistical modeling using Python and SQL. Strong foundation in exploratory data analysis, feature engineering, and predictive modeling on real-world datasets. Experienced in transforming raw data into actionable insights through structured analytical workflows. Effective team collaborator with the ability to communicate data-driven insights clearly.

EXPERIENCE

Data Science Intern – AB Technologies

 Feb 2023 – Mar 2023

 Nagercoil, India

Built and optimized data analysis and machine learning workflows on large structured datasets.

- Built end-to-end data pipelines using Python, SQL, and Pandas to ingest, clean, and analyze 100K+ structured records, supporting data analysis, reporting, and machine learning workflows.
- Performed exploratory data analysis (EDA), data preprocessing, and feature engineering, including data validation, missing value treatment, outlier handling, encoding, and normalization, to improve data quality and analytical accuracy.
- Developed and optimized machine learning and deep learning models using scikit-learn and neural network techniques, applying cross-validation, hyperparameter tuning, and performance evaluation, resulting in 50%+ improvement in model accuracy.
- Created data visualizations, dashboards, and analytical reports using Tableau, Matplotlib, and Seaborn to communicate KPIs, trends, and insights to stakeholders and enable data-driven decision making.

Volunteer Tutor – RCET

 Sep 2022 – Mar 2023

 Kanyakumari, India

Delivered hands-on training in data analysis and machine learning fundamentals to support student skill development.

- Delivered structured training in Python, data analysis, statistics, and machine learning fundamentals to 15+ students, combining theoretical concepts with hands-on coding using real-world datasets.
- Conducted 50+ interactive problem-solving and lab sessions, strengthening students' data interpretation, analytical reasoning, and model-building skills.
- Designed customized learning plans, practice exercises, and mini-projects to improve concept retention, technical confidence, and practical application of core data science concepts.

EDUCATION

Masters in Data Science
Universitaet Koblenz

 Oct 2024 – Present

Bachelors in Computer Science
Anna University | CGPA: 1.7

 Aug 2019 – Jul 2023

CERTIFICATIONS

- Google Data Analytics Certificate: Coursera 
- Data Science Bootcamp: Udemy 
- Machine Learning with Python: Udemy 
- Deep Learning Specialization: DeepLearning.AI 
- SQL for Data Analytics: Udemy 

SKILLS

Python

SQL

NoSQL

Data Analysis

Data Science

Statistical Analysis

Tableau

Hypothesis Testing

Power BI

Excel

Exploratory Data Analysis

Machine Learning

Deep Learning

Regression Models

scikit-learn

Classification Models

TensorFlow

Pandas

Numpy

Matplotlib

Seaborn

Feature Engineering

MySQL

MongoDB

KPI Analysis

Git

Docker

PySpark

AWS

M365

Data Cleaning

Business Insights

Reporting

RESEARCH EXPERIENCE

Research Assistant (AI & Data Science) – Universität Koblenz

📅 Oct 2025 – Present

- Conducted a comparative study of machine learning and deep learning models for structured tabular classification and multivariate time series forecasting using an end-to-end modeling workflow.
- Implemented and evaluated classification and forecasting models (Logistic Regression, SVM, Random Forest, XGBoost, Neural Networks), achieving up to ~83% accuracy, ~0.89 ROC-AUC, and 20–30% lower RMSE for short-term forecasts.
- **Skills:** Machine Learning, Deep Learning, Data Analysis, Time Series, scikit-learn, Tensorflow, Pandas, NumPy, Statistics, Matplotlib, Seaborn, etc.

PROJECTS

Hospital Admission Analytics & Predictive Modeling 🔗

- Built machine learning models in Python to forecast hospital admissions using structured healthcare data, enabling data-driven capacity planning and operational insights.

Real Estate Fair Value & Overpricing Predictor 🔗

- Developed regression-based ML models to estimate fair property values and identify overpricing patterns from real estate datasets.

Workplace Safety Risk Intelligence (Deep Learning) 🔗

- Designed a deep learning–based risk prediction system using TensorFlow to assess and classify workplace safety hazards.

PySpark-Based Fraud Detection System 🔗

- Implemented a scalable fraud detection pipeline using PySpark and gradient boosting models on large-scale transactional data.

Tableau Dashboard – World’s Finest Restaurants 🔗

- Created interactive Tableau dashboards to analyze global restaurant trends, ratings, and geographic insights.

Bayut Property Sales Intelligence (Tableau) 🔗

- Built Tableau-based analytics dashboards to visualize property sales trends and pricing behavior for real estate insights.

NoSQL-Based Music Data Analytics (MongoDB) 🔗

- Designed a NoSQL analytics framework using MongoDB to store and analyze large-scale music datasets efficiently.

Statistical Analysis of International T20 Cricket Performance 🔗

- Performed statistical analysis and visualization in Python to evaluate and compare international T20 cricket batting performance.

COMPETENCIES

- Data analysis, validation and quality assurance
- Data cleaning, preprocessing, and transformation workflows
- Exploratory data analysis (EDA) and statistical interpretation
- Feature engineering and dataset preparation for modeling
- Predictive analytics, supervised machine learning solutions
- Model evaluation, performance metrics, and validation techniques
- End-to-end model development and deployment using Streamlit for analytical applications
- Business intelligence reporting and dashboard development
- Data storytelling and stakeholder-oriented reporting
- Business intelligence and data visualization tools
- Working knowledge of relational and NoSQL database systems
- Applied statistics for data-driven decision making

LANGUAGES

English — C1 (Advanced)

German — Beginner

ACHIEVEMENTS

- **Academic Topper Award recipient (3+ times)**, ranking among the top academic performers, graduating with a **CGPA of 9.0/10** and awarded **First Class with Distinction**.
- Winner and finalist in inter-college and national-level technical competitions.
- Served as a **council member**, contributing to academic and technical event coordination and team initiatives.

STRENGTHS

Analytical thinking	Problem solving
Data-driven mindset	Adaptability
Continuous learning	Teamwork
Attention to detail	Critical thinking